

In the 9th century, the Muslim scholar al-Khwarizmi developed the Hindu-Arabic number system and made important advances in algebra. These innovations underpinned the subsequent development of trade, technology, and science.

MILESTONES

EXCELS IN ACADEMIA

Becomes director of the House of Wisdom under Caliph Al-Ma'mun in the early 800s CE.

CARTOGRAPHIC WORK

Helps compile a world map; leads a project to measure the circumference of Earth between c.811 and 825 CE.

TRANSFORMS MATH

Writes a book on numerals in c.825 CE; its Latinized title is the origin of the word "algorithm."

ADVANCES ALGEBRA

Publishes the world's first algebra book in c.830 CE; it includes the theory of quadratic equations.

During his life's work at the House of Wisdom, a center of scientific research and teaching in Baghdad (now in Iraq), Muhammad ibn Musa al-Khwarizmi wrote influential treatises on cartography and astronomy, along with two books that had profound consequences for Arabic and Western mathematics. Adapting an existing Indian numbering practice, he proposed the figures 0 and 1 to 9 as the basis of a new number system and showed how to do arithmetic with them. This system, which became known as Hindu-Arabic numerals, was introduced to Europe in the 12th century through a Latin translation of the book *Al-Khwarizmi Concerning the Hindu Art of Reckoning*. Today, it has been adopted almost everywhere.

Father of algebra

Drawing on the work of ancient mathematicians, al-Khwarizmi wrote the foundational textbook on algebra. His book gave the first systematic explanation of the symbols and methods used to solve equations. For example, he demonstrated how to work out numerical problems by transposing elements from one side of an equation to the other and canceling out items that appear on both sides. Al-Khwarizmi's work on arithmetic and algebra was built on by later mathematicians and went on to transform the way people did business by simplifying financial transactions and enabling accurate accounting. It also helped scientists hone their theories about the natural world.

One of al-Khwarizmi's achievements

was to produce a map of the river Nile. It shows the river flowing from a place called Moon Mountain (right) to the Mediterranean Sea (left).

